The EU Carbon Border Adjustment Mechanism and China's Strategic Response: Carbon Pricing Realignment and Industrial Decarbonization

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Abstract—The European Union's Carbon Border Adjustment Mechanism (CBAM), entering its definitive phase in January 2026, operationalizes Nordhaus's climate club logic by linking market access to carbon pricing. This paper examines how CBAM reshapes China's carbon pricing architecture and sectoral decarbonization in steel and cement, the two sectors driving China's CBAM exposure. Using plant-level data from the Global Energy Monitor trackers (458 steel and 1,016 cement plants) and a liability model derived from Regulation EU 2023/956, we identify three transmission channels: a direct cost channel, a revenue retention incentive, and a signalling channel benchmarking CN-ETS ambition against EU ETS prices. At the prevailing price gap, BF-BOF steel exports face a net liability of EUR 140.7/t, versus EUR 6.7/t for EAF steel and EUR 42.1/t for cement—a 17.6:1 intensity ratio that maps directly onto competitiveness exposure. CN-ETS reforms announced since 2023—absolute caps by 2027, sectoral expansion, and gradual auctioning—correlate with, and are plausibly reinforced by, CBAM's fiscal logic, though domestic dual-carbon goals provide independent momentum and causal attribution lies beyond this study's scope. We contribute an empirically calibrated assessment of CBAM's first-year implementation and identify conditions under which trade-linked enforcement complements rather than substitutes for multilateral cooperation.

Keywords: CBAM; CN-ETS; carbon pricing convergence; industrial decarbonization

I. Introduction

A. *From Free-Riding to Trade-Linked Enforcement: Why CBAM Matters*

Global climate governance has long suffered from the tragedy of the atmospheric commons: mitigation benefits are universally shared, while abatement costs fall exclusively on individual jurisdictions Hardin (1968). This structural free-riding incentive has undermined every multilateral architecture, from the Kyoto Protocol's unenforceable targets to the Paris Agreement's voluntary NDCs. Nordhaus's Nordhaus (2015) climate club proposal addresses this enforcement gap by proposing a coalition of high-ambition jurisdictions that adopt a common internal carbon price and impose uniform border tariffs on non-members, making free-riding costlier than abatement.

For nearly a decade, the climate club remained a theoretical construct. This changed with the CBAM, which entered its transitional phase in October 2023 and definitive phase in January 2026—the first instrument to operationalize trade-linked carbon enforcement at scale European Union (2023). Under Regulation EU 2023/956, importers of six carbon-intensive goods must surrender certificates priced at the weekly average EU ETS allowance price. Critically, verified carbon costs paid in the country of origin are deducted euro for euro from CBAM liability. This deduction clause is CBAM's strategic core: it transforms cross-border price differentials into a direct fiscal incentive for non-EU jurisdictions to strengthen domestic carbon pricing and retain revenue locally.

B. *CBAM's Particular Relevance for China*

As the EU's largest non-EU source of CBAM-liable imports, China represents the most consequential test case for the climate club model. Three factors define this relationship.

First, the carbon price gap is large and policy-actionable. In 2024–2025, EU ETS allowances traded at EUR 50–75/tCO_2, while CN-ETS allowances cleared at EUR 10–13/tCO_2 Ministry of Ecology and Environment of China (2023). At an EU ETS price of EUR 80/tCO_2, steel produced via the blast furnace–basic oxygen furnace (BF-BOF) route faces a net liability of EUR 140.7/t, compared to just EUR 6.7/t for electric arc furnace (EAF) steel—a 17.6:1 intensity ratio that directly maps emissions performance to competitiveness.

Second, China's industrial structure creates structural vulnerability. Its steel sector (458 plants, 1,005 Mtpa—million tonnes per annum—capacity, 54% of global total) is 83.1% BF-BOF based, compared to 44% in the EU and 29% globally World Steel Association (2024). Its cement industry (1,016 plants, 1,772 Mtpa capacity) has a clinker-to-cement ratio of 0.760, exceeding the global average of 0.73. The carbon-intensive technology mix that enabled China's industrialization is precisely what CBAM penalizes most heavily.

Third, CBAM's timeline aligns with a critical domestic reform window. Between 2023 and 2027, China's Ministry of Ecology and Environment announced three landmark CN-ETS reforms: transition to absolute emissions caps by 2027, expansion to steel and cement (2024) and aluminium (2025–2026), and gradual introduction of allowance auctioning Ministry of Ecology and Environment of China (2021, 2023). While China's "dual carbon" goals provide independent momentum, the temporal alignment with CBAM invites scrutiny of how external pressure interacts with domestic reform—an analytical task this paper takes up.

C. Research Question, Scope, and Contribution

This paper addresses the central question: *How does CBAM reshape China's carbon pricing architecture and sectoral decarbonization pathways?* We decompose this into three sub-questions: (i) what channels transmit CBAM pressure to Chinese policymakers and industry; (ii) what is the magnitude of sectoral cost exposure under current and reformed CN-ETS scenarios; and (iii) what political-economy constraints bound feasible reform trajectories.

Our empirical analysis focuses on steel and cement during 2024–2027. Aluminium is excluded from quantitative analysis due to limited plant-level data but discussed qualitatively where the policy logic generalizes. Methodologically, we combine Global Energy Monitor plant-level data with a CBAM liability model derived directly from Regulation EU 2023/956.

This paper makes three contributions: (i) an empirically calibrated assessment of CBAM's first-year implementation, moving beyond pre-implementation modelling; (ii) a three-channel transmission framework linking CBAM design to CN-ETS reform; and (iii) a clarification of the climate club model's limits in non-OECD contexts.

The remainder of the paper proceeds as follows. Section II reviews the literature and identifies gaps. Section III describes data and methods. Section IV analyses CBAM design and transmission channels. Section V presents empirical results. Section VI examines industrial and policy responses. Section VII discusses implications. Section VIII concludes.

II. Literature Review and Theoretical Framework

A. Climate Clubs and the CBAM as a De Facto Club

Nordhaus's Nordhaus (2015) climate club model proposes that a coalition adopting a common internal carbon price and imposing *uniform* penalty tariffs on non-members can sustain stable high-abatement coalitions. A growing literature examines whether the EU's CBAM operationalises this logic. Tagliapietra and Wolff Tagliapietra and Wolff (2021) and Szulecki et al. Szulecki et al. (2022) characterise CBAM as a "de facto" climate club, yet emphasise that its incentive structure diverges from Nordhaus's stylised model: rather than excluding non-members through uniform tariffs, CBAM's deduction clause creates a fiscal pull toward domestic carbon pricing. More fundamental critiques question whether the club mechanism captures the actual politics of climate cooperation. Falkner Falkner (2016) and Falkner, Nasiritousi, and Reischl Falkner et al. (2022) argue that minilateral clubs face severe legitimacy and feasibility constraints, and that "the most stringent climate club proposals are focused on overcoming the free-riding problem rather than addressing distributive conflict" Falkner et al. (2022). Aklin and Mildenberger Aklin and Mildenberger (2020) push this further, arguing that climate inaction reflects domestic distributive conflict, not international free-riding—governments adopt climate policies regardless of others' actions, suggesting external sanctions target the wrong binding constraint. Böhringer et al. Böhringer et al. (2022) and Mehling et al. Mehling et al. (2019) converge on a similar diagnosis from a design perspective: real-world border adjustments must accommodate vastly heterogeneous national circumstances, generating administrative and equivalence challenges that the homogenous-region Nordhaus framework abstracts away. This paper tests whether these heterogeneities undermine the climate club's self-enforcing logic in the Chinese context.

B. Two-Level Games in Authoritarian Climate Governance

Putnam's Putnam (1988) two-level game framework, in which negotiators simultaneously bargain at international and domestic tables constrained by overlapping "win-sets," has become a standard lens for analysing climate diplomacy Bailer (2012). Its application to China requires substantial adaptation. The original framework presumes electoral ratification, whereas China's domestic "Level II" game is constructed through bureaucratic competition among ministries (MEE, NDRC, MIIT, MOFCOM), provincial fiscal interests, and state-owned enterprise lobbying Feng (2021); Gilley (2012). Recent work by Chen and Lo Chen and Lo (2025) on "authoritarian environmentalism 2.0" demonstrates that market-based instruments such as ETS reinforce hierarchical state control rather than constraining it—suggesting that international pressure interacts with Chinese policy through intra-state distributive bargaining rather than legislative veto. This reframing is consequential: external instruments such as CBAM enter Chinese policy debates not by changing aggregate national costs but by reshaping the relative bargaining position of pro-reform actors within the bureaucracy. We operationalise this logic by examining how CBAM's differential cost impacts across technologies and provinces alter the domestic win-set for CN-ETS reform.

C. Carbon Pricing Convergence and ETS Diffusion

The literature on ETS diffusion offers two contending expectations. The optimistic view—underlying linkage proposals from the EU–Switzerland model—anticipates convergence of design as jurisdictions emulate the EU template. The empirical record is more sobering: Wettestad and Gulbrandsen Wettestad and Gulbrandsen (2018) document that ETS diffusion has produced design *divergence* more often than convergence, as local political-economy factors override technocratic emulation. China's case illustrates this divergence. Stoerk, Dudek, and Yang Stoerk et al. (2019) warned at CN-ETS launch that intensity-based caps with ex-post adjustment could undermine price signals—a critique now being addressed through the announced 2027 transition to absolute caps. Cui et al. Cui et al. (2021) provide firm-level evidence that even China's regional pilots reduced emissions by 16.7% without significant output loss, suggesting that low headline prices need not imply policy ineffectiveness. Dai and Pollitt Dai and Pollitt (2025), drawing on extensive document and interview evidence, conclude that CN-ETS expansion is constrained primarily by MRV bottlenecks rather than political will, and that CBAM imposes an effective 2034 deadline for full CN-ETS maturity. Together, these studies suggest that CBAM does not produce convergence directly; it alters the political-economy calculus of jurisdictions whose pre-existing reform trajectories were already underway. This paper quantifies the magnitude of this "accelerator effect" by comparing CN-ETS reform timelines before and after CBAM's announcement.

D. Empirical Evidence on CBAM's Post-Implementation Effects

Until 2024, almost all CBAM impact assessments relied on pre-implementation modelling. Post-implementation empirical work remains sparse but is rapidly accumulating, with two methodologically distinct studies forming the baseline for our analysis. Li et al. Li et al. (2026), using a global multi-regional input–output (GMRIO) framework, estimate that China's CBAM-induced export losses rise from EUR 0.78 billion under current scope to EUR 11.14 billion under full expansion, and find that CN-ETS can offset 30–60% of these losses depending on price scenarios and CBAM coverage. Pan et al. Pan et al. (2025) couple GMRIO with the GCAM-China integrated assessment model at provincial resolution and report offset rates of 42% (DICE model) to 49% (ECE scenario) for iron, steel, aluminium, and products (ISAP) exports specifically, with over 65% of provincial costs concentrated in Hebei, Jiangsu, Shanxi, Liaoning, and Zhejiang. The two estimates fall within overlapping ranges, but the divergence in headline figures reflects substantively different assumptions: Li covers all CBAM sectors with static GMRIO and price-scenario variation, while Pan focuses on ISAP with dynamic modelling that captures EAF technology penetration as a mitigation channel. The two studies are best read as complementary rather than competing, but neither traces the political-economy mechanism by which CBAM pressure becomes domestic CN-ETS reform.

Complementary evidence reinforces the picture: Magacho, Espagne, and Godin Magacho et al. (2024) establish the heterogeneity of CBAM exposure across EU trade partners; Wang et al. Wang et al. (2024) estimate that CBAM reduces Chinese steel exporters' profits by 58% even as environmental benefits remain marginal at the sectoral level; Clausing et al. Clausing et al. (2025) document empirical instances of CBAM-induced policy diffusion, including China's March 2025 expansion of CN-ETS to CBAM-covered sectors. The divergence between Li et al. Li et al. (2026) and Pan et al. Pan et al. (2025) highlights a broader methodological limitation: static CGE models capture aggregate cost exposures but miss dynamic technology adjustment, while dynamic IAMs incorporate mitigation channels but rely on sector-average emission factors. This paper addresses both limitations by using plant-level data to model technology heterogeneity within a static liability framework calibrated to post-implementation policy parameters.

E. Research Gaps and Analytical Framework

Synthesising the above literature, we identify three critical departures of CBAM from Nordhaus's stylised climate club model that have not been jointly theorised or empirically tested (summarised in Fig. 1). *First*, CBAM's deduction clause replaces Nordhaus's uniform tariff with a differentiated, emission-intensity-dependent liability that endogenises the choice of domestic carbon pricing—transforming the club mechanism from exclusionary tariff to gravitational fiscal pull. *Second*, CBAM's non-member targets exhibit vast industrial-structure heterogeneity that Nordhaus's homogenous-region C-DICE framework cannot capture; China's BF-BOF dominance and provincial concentration of CBAM exposure Pan et al. (2025) generate response capacities and constraints fundamentally different from those of EU member states. *Third*, the binding constraint on CN-ETS reform is not international free-riding but domestic distributive conflict among provincial governments, SOEs, and central reformers Aklin and Mildenberger (2020); Cui et al. (2021); Dai and Pollitt (2025)—a constraint Nordhaus's external-enforcement logic does not address.

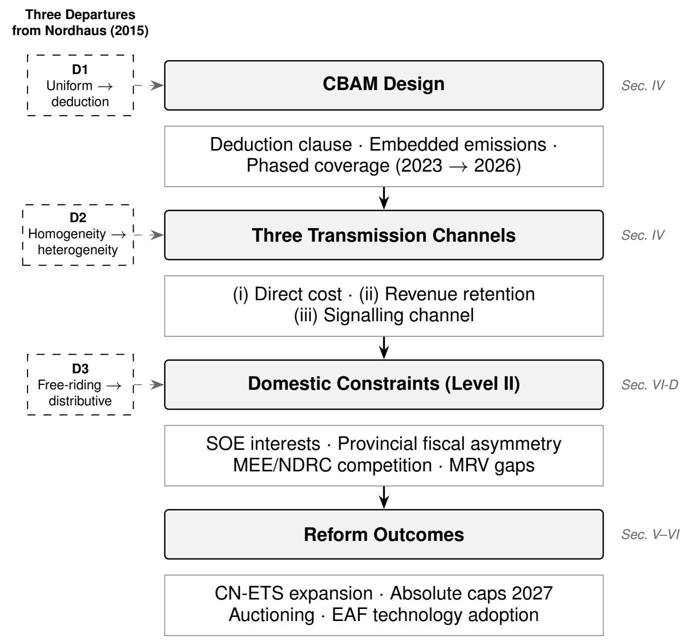

Fig. 1. Analytical framework. CBAM's design (Stage 1) generates three transmission channels (Stage 2) that interact with China's domestic political-economy constraints (Stage 3) to shape CN-ETS reform outcomes (Stage 4). Three departures from Nordhaus's stylised climate club (left, D1–D3) define the paper's theoretical contribution; Stage 3 constraints operationalise Putnam's Level II.

These departures motivate three core research gaps that this paper addresses:

1) **Mechanism gap**: Existing CGE and IAM studies compute equilibrium offset outcomes without mapping the causal transmission mechanism from CBAM external pressure to specific CN-ETS reform measures.
2) **Heterogeneity gap**: Sector-average emissions modelling obscures the plant-level technology heterogeneity that drives the BF-BOF–EAF cost differential central to China's adjustment pathway.
3) **Context gap**: No study has systematically integrated domestic distributive conflict into an empirical assessment of CBAM's impact on non-OECD carbon pricing

systems.

To address these gaps, we develop a transmission-channel framework comprising the direct cost channel, the revenue retention incentive, and the signalling channel (Section IV), and apply it to plant-level data on China's steel and cement sectors (Sections V–VI). The framework operationalises Putnam's two-level logic in the authoritarian-bureaucratic context and integrates the distributive-conflict reading of climate politics into a tractable empirical assessment.

III. Methodology and Data

A. Data Sources

Analysis draws on four primary datasets. Plant-level steel capacity and technology data come from the GEM Iron & Steel Tracker (Global Energy Monitor, 2026), comprising 458 operating plants with unit-level BOF/EAF technology classification, provincial location, and production capacity. Cement capacity data come from the GEM Cement and Concrete Tracker (Global Energy Monitor, 2025), covering 1,016 operating plants with clinker and cement capacity disaggregated by plant type (integrated vs. grinding-only). Sectoral production benchmarks draw on International Energy Agency (2023) and World Steel Association (2024). Carbon pricing data for the CN-ETS are sourced from MEE official publications (Ministry of Ecology and Environment of China, 2021, 2023); EU ETS price ranges reflect 2024–2025 market data.

B. Emission Factor Derivation

Three sector-specific emission factors are applied throughout:

BF-BOF steel: 2.10 tCO_2/t crude steel, representing Scope 1 direct process emissions per IEA benchmarks (International Energy Agency, 2023), consistent with near-exclusive reliance on coking coal as blast furnace reductant across China's 599 operating blast furnaces.

EAF steel: 0.10 tCO_2/t crude steel (Scope 1, **CBAM-applicable only**). Under Regulation EU 2023/956, Scope 2 (indirect) electricity emissions are **excluded** from CBAM certificate liability for steel; only Scope 1 direct process emissions apply. EAF Scope 1 of 0.10 tCO_2/t reflects direct emissions from DRI and pig iron inputs (International Energy Agency, 2023). For reference, total EAF emissions intensity including Scope 2 is 0.44 tCO_2/t: 580 kWh/t electricity consumption at the MEE national grid factor of 0.581 $kgCO_2$/kWh (Ministry of Ecology and Environment of China, 2023) yields 0.337 tCO_2/t Scope 2, plus 0.10 tCO_2/t Scope 1. This total figure (0.44) is reported for technology comparison only and does *not* affect CBAM liability. China's power generation mix and the derivation of this grid factor are shown in Fig. 2.

Cement: 0.627 tCO_2/t finished cement, derived from the GEM-observed clinker-to-cement ratio of 0.760 multiplied by the IPCC (2006) clinker emission factor of 0.825 tCO_2/t clinker (calcination: 0.525; coal combustion kiln heat: 0.300) (Intergovernmental Panel on Climate Change, 2006).

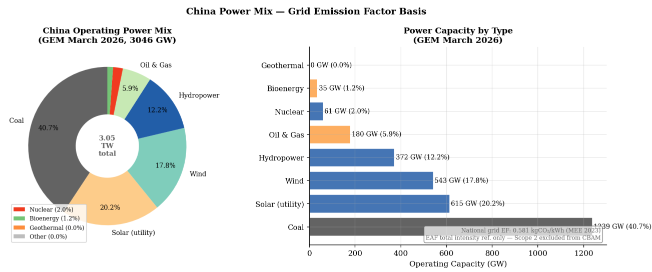

Fig. 2. China's power generation mix and associated grid emission factor (0.581 $kgCO_2$/kWh) (Ministry of Ecology and Environment of China, 2023). This underpins EAF total emissions intensity (0.44 tCO_2/t including Scope 2); Scope 2 is excluded from CBAM liability for steel under Regulation EU 2023/956 (Scope 1 only: 0.10 tCO_2/t).

C. CBAM Liability Model

Net CBAM liability per tonne of exported product i is computed as:

$$\text{CBAM Liability}_i = \max\Big(0,\ EF_i \times (P_{\text{EU}} - P_{\text{CN}})\Big) \qquad (1)$$

where EF_i is embedded emissions intensity (tCO_2/t), P_{EU} is the prevailing EU ETS allowance price (EUR/tCO_2), and P_{CN} is the verified CN-ETS carbon cost paid in the country of origin. Steel and cement sectors entered CN-ETS coverage in 2024; P_{CN} therefore reflects the prevailing CN-ETS allowance price for covered facilities, currently EUR 10–13/tCO_2. Sensitivity analysis spans two EU ETS price scenarios (EUR 65 and EUR 80/tCO_2) and five CN-ETS price states.

D. Scope and Limitations

Aluminium is excluded from quantitative analysis due to the absence of GEM unit-level smelting data for China. Aggregate CBAM exposure estimates are bounded to current CBAM sectoral scope (HS Chapter 72 iron and steel; cement clinker); downstream embedded emissions in manufactured goods are noted but not modelled. Provincial export volumes are approximated from capacity shares rather than trade flow data, introducing uncertainty in absolute cost estimates.

Causality is a further limitation: the correlation between CBAM implementation and CN-ETS reform acceleration is consistent with the analytical framework in Section II.E, but formal causal identification requires counterfactual evidence beyond the scope of this study. Findings are interpreted as associations informed by theoretical priors, not demonstrated causal effects. These parameters feed directly into the liability model applied in Section V.

IV. CBAM Design and Transmission Mechanisms

A. Policy Design and Transmission Logic

CBAM (Regulation EU 2023/956) entered its definitive phase on 1 January 2026, following a transitional reporting-only phase from 1 October 2023 through 31 December 2025 (European Union, 2023). EU importers of products listed in Annex I—iron and steel, aluminium, cement, fertilisers, electricity, and hydrogen—must now register as authorised CBAM declarants, surrender CBAM certificates by 31 May each year against the preceding calendar year's verified embedded emissions, and reconcile any outstanding liability with EU customs. Certificate prices track the weekly auction-clearing price of EU ETS allowances, ensuring price equivalence between domestic compliance and border-adjusted imports.

Embedded emissions are defined as Scope 1 direct emissions for steel, aluminium, and hydrogen, with Scope 2 indirect electricity-related emissions added for cement, fertilisers, and—through the 2026 sectoral review—potentially aluminium. Where verified installation-level data are unavailable, default values published by the European Commission apply; these are typically benchmarked at the worst-decile producers globally, generating a strong incentive for facility-level monitoring, reporting, and verification (MRV).

CBAM's strategic significance for China rests on Article 9 of the Regulation, which permits the deduction of carbon costs already paid in the country of origin from CBAM liability on a euro-for-euro basis, provided those costs are verifiable and meet equivalence criteria specified in implementing regulations. This deduction clause is the fulcrum of the mechanism's interaction with Chinese carbon policy: each yuan of effective CN-ETS compliance cost converts what would otherwise be a transfer to the EU budget into revenue retained domestically. The parallel free-allocation phase-out under EU ETS Annex I—running 2026–2034 in linear annual decrements—calibrates CBAM coverage to internal carbon-cost equalisation rather than additional protection, a design feature critical to WTO defensibility.

Three transmission channels follow from this architecture and are tested empirically in Sections V and VI:

- The **direct cost channel**: CBAM raises the landed cost of Chinese exports in proportion to embedded emissions intensity, with the per-tonne liability given by Eq. 1. The competitiveness implication scales with the Chinese carbon-intensity disadvantage relative to EU producers operating under stringent ETS prices.
- The **revenue retention incentive**: the Article 9 deduction transforms domestic carbon pricing from a pure compliance cost into a fiscal-substitution instrument. Each marginal increase in P_{CN} shifts an equivalent quantum of revenue from EU coffers to Chinese ones, bounded above by P_{EU}. This converts a passive trade-policy exposure into an active strategic incentive for CN-ETS reform.
- The **signalling channel**: the publicly observable P_{EU} functions as an external benchmark for CN-ETS ambition. Even where direct deduction logic is incomplete (e.g., during sectoral phase-in delays), the price gap is read by domestic policy actors—MEE, NDRC, sectoral associations—as an indicator of how far the reform agenda must travel.

A fourth, procedural channel emerges indirectly from MRV requirements: CBAM's documentation standards effectively impose ISO 14064-aligned carbon accounting on Chinese exporters as a precondition for accessing facility-specific (rather than punitive default) emission factors. From the firm's perspective this is sunk compliance cost; from the system's perspective it is a public-good investment in carbon-market infrastructure that complements CN-ETS expansion—an interaction Dai and Pollitt (2025) document as a binding bottleneck on CN-ETS maturity.

B. Comparative Architecture: EU ETS and CN-ETS

CBAM's revenue-retention logic operates entirely through the price gap between the two emissions trading systems on either side of the Article 9 deduction. Understanding the structural drivers of that gap requires direct comparison of EU ETS and CN-ETS architectures, summarised in Table I.

The EUR 40–65 / tCO_2 price gap is structurally generated by three differences. *First*, EU ETS imposes an absolute cap whose annual contraction is mechanically priced into the forward curve, whereas CN-ETS intensity benchmarking allows allowance volumes to expand with output, dampening upward price pressure (Dai and Pollitt, 2025; Stoerk et al., 2019). *Second*, EU allocation has been steadily migrating from free to auctioned, raising effective compliance costs in line with the cap, while Chinese steel and cement entered CN-ETS in 2024 with generous free allocation to ease the transition. *Third*, EU ETS coverage spans the bulk of stationary-combustion and industrial emissions, while CN-ETS covered only the power sector through 2023 and remains in mid-expansion.

These structural features are the binding constraint on convergence. Within the two-level games framework adopted in Section II-B, they define the upper limit of how rapidly CN-ETS can be tightened without compromising sectoral employment and energy-security imperatives (Nahm and Urpelainen, 2021). CBAM enters this domestic equilibrium as an external incentive that shifts the cost–benefit calculation of reform—not by mandating convergence but by attaching a measurable fiscal-leakage cost to non-convergence.

C. Sectoral Coverage Logic and Expansion Trajectory

The sectors selected for CBAM Annex I coverage—cement, iron and steel, aluminium, fertilisers, electricity, and hydrogen—share three features that define the mechanism's analytical logic. *First*, each is highly carbon-intensive, with embedded emissions in the range of 0.6–2.1 tCO_2 per tonne of finished product. *Second*, each has documented exposure to carbon leakage under the EU ETS prior to CBAM, evidenced by free-allocation eligibility and trade-intensity metrics maintained by the European Commission. *Third*, each is technologically mature enough to permit reasonably standardised emission-factor methodologies, securing administrative feasibility within a customs-based system.

Hydrogen and electricity are forward-looking inclusions. Hydrogen exposure is currently small but anticipated to grow as the EU's Renewable Energy Directive recast (RED III) drives demand for low-carbon hydrogen imports; electricity inclusion addresses cross-border grid leakage from non-EU jurisdictions adjacent to EU member states (Western Balkans, Türkiye), and is of limited direct relevance for China.

TABLE I
STRUCTURAL COMPARISON OF EU ETS AND CN-ETS (2024–2025)

Dimension	**EU ETS**	**CN-ETS**
Year established	2005 (Phase I); Phase IV running 2021–2030	2021 (national); regional pilots from 2013
Sectoral coverage	Power, energy-intensive industry, intra-EEA aviation, maritime (from 2024)	Power generation; steel and cement (from 2024); aluminium planned 2025–26
Cap design	Absolute, declining; Linear Reduction Factor 4.3% (2024–27), 4.4% (2028–30)	Intensity-based; transition to absolute cap announced for 2027
Allocation method	Hybrid: auctioning ($\sim$57%); free benchmark-based allocation declining via cross-sectoral correction factor	Free allocation against intensity benchmarks; auctioning pilots in select provinces
Price (2024–25)	EUR 50–80 / tCO_2	EUR 10–13 / tCO_2 (CNY 80–105)
MRV regime	Accredited third-party verifiers under ISO 14065; central registry	Sectoral verifiers under MEE supervision; provincial registries integrating into national
Compliance penalty	EUR 100/tCO_2 + missed-allowance surrender + name disclosure	Administrative penalty typically 5–10$\times$ market price; reform pending
Linkage capability	Linked with Swiss ETS since 2020	No active linkage; bilateral cooperation under exploration
Revenue use	Member states (climate purposes $\geq$50%); Innovation Fund; Modernisation Fund	MEE; sectoral allocations; reform direction toward general fiscal absorption
Offsets	Limited international credits, no longer accepted post-Phase III	CCER (China Certified Emission Reductions) reactivated 2024, capped

The phase-in trajectory is designed for parallelism with EU ETS free-allocation reduction. Free allowances for CBAM-covered EU sectors decline from 100% in 2025 to 0% in 2034 in linear annual decrements; CBAM certificate obligations rise correspondingly, ensuring that European producers always face the full carbon cost of their emissions and that Chinese exporters face an equivalent cost on the residual gap not covered by CN-ETS payments. This calibration is critical both to WTO defensibility and to the mechanism's framing as equalisation rather than protection.

Two expansion vectors are particularly consequential for China. The first is the 2025–26 mandated review of CBAM scope, widely expected to recommend extension to organic chemicals, polymers, and selected downstream manufactured goods (refined steel products, aluminium semi-fabricates, cement-based construction materials). Li et al. (2026) project that this expansion would lift China's aggregate CBAM exposure from EUR 0.78 billion under current pilot scope to EUR 11.14 billion under full coverage—a 14$\times$ increase reflecting both higher product volumes and the deeper integration of CBAM-regulated inputs into Chinese manufacturing exports. The second vector is the expansion of indirect (Scope 2) emissions coverage. Currently included only for cement and fertilisers, the inclusion of Scope 2 for steel and aluminium would substantially raise the per-tonne liability of Chinese producers reliant on coal-dominant grids; the elasticity of this exposure to provincial grid factors is quantified in Section V-D.

D. Strategic Foresight: CBAM as a Precedent for Global Border Carbon Adjustments

CBAM is no longer a single-jurisdiction risk for Chinese exporters. The United Kingdom has confirmed that its own CBAM will enter force on 1 January 2027, covering aluminium, cement, fertilisers, hydrogen, and iron and steel under primary legislation enacted in the Finance Act 2026, with secondary legislation under technical consultation through mid-2026 (HM Revenue and Customs, 2026). The UK design parallels the EU regime but excludes electricity and operates with a single global default value per CN code in its first year, simplifying administration at the cost of country-of-origin differentiation. The UK and EU concluded an agreement in principle in May 2025 to link their respective ETS systems and explore mutual CBAM exemption, which would consolidate a transatlantic European carbon-pricing zone with implications for Chinese MRV-equivalence negotiations.

In the United States, the bipartisan Foreign Pollution Fee Act of 2025 (S. 1325) was reintroduced in April 2025 by Senators Cassidy and Graham, proposing tiered tariffs on imports of iron, steel, aluminium, cement, glass, fertilisers, hydrogen, and selected critical-mineral and battery inputs (Cassidy and Graham, 2025). Unlike the EU and UK mechanisms, the US proposal does not impose a domestic carbon price and does not formally credit foreign carbon costs; it computes pollution-intensity differentials relative to US production and applies ad valorem tariffs accordingly. This design departs from the price-equivalence logic of EU CBAM and would, if enacted, expose Chinese producers to a parallel charge with no offset pathway through CN-ETS payments. Whether the bill advances is uncertain—prior versions stalled in 2023–24—but its bipartisan support and explicit geopolitical framing against China give it material policy weight (Center for Strategic and International Studies, 2024). The accompanying PROVE IT Act, which would direct the U.S. Department of Energy to study comparative carbon intensities, advanced from committee in the 118th Congress and provides the data infrastructure on which any subsequent border-adjustment regime would rely.

Canada's federal climate plan under the Carney government has indicated active consideration of a CBAM as part of its 2030 industrial-policy package, and Australia conducted public consultations on a Carbon Leakage Review concluding in 2024. While neither jurisdiction has confirmed implementation, the cumulative direction of policy diffusion is unambiguous: the EU's first-mover act has shifted the equilibrium of acceptable trade–climate policy, and China should plan for cumulative exposure to multiple, partially overlapping border-adjustment regimes by 2028–30 rather than for the EU mechanism in isolation.

This diffusion has two strategic implications. First, the marginal value of strengthening CN-ETS rises with each additional jurisdiction adopting deduction-based CBAMs, since the same domestic carbon cost is increasingly offsettable across multiple borders. Second, the heterogeneity of mechanism designs—deduction-based (EU, UK) versus differential-based (US)—means that no single CN-ETS reform pathway optimises against all of them simultaneously; harmonisation pressures in MRV standards and verifier accreditation will likely intensify, whether through bilateral mutual-recognition agreements or multilateral fora such as the OECD Inclusive Forum on Carbon Mitigation Approaches (IFCMA).

V. Empirical Results

A. Sectoral Exposure: China's CBAM Vulnerability

China constitutes the single largest non-EU source country for CBAM-liable exports, with exposure concentrated in steel and cement.

Steel. China's GEM-tracked steel industry comprises 458 plants with 1,005 Mtpa of operating crude steel production capacity—approximately 54% of global crude steel production (World Steel Association, 2024). Of this capacity, 835.7 Mtpa (83.1%) is attributable to BF-BOF operations and 169.5 Mtpa (16.9%) to EAF. China's EAF share relative to global peers is shown in Fig. 3. China's weighted-average CBAM-applicable emissions intensity (Scope 1) across its current production mix is $0.831 \times 2.10 + 0.169 \times 0.10 = 1.76$ tCO_2/t, vs. 0.10 tCO_2/t for a pure EAF producer—a 17.6:1 ratio that translates directly into the core CBAM competitiveness liability.

Li et al. (2026) estimate aggregate CBAM-induced export losses rising from EUR 0.78 billion under the current pilot sectors to EUR 11.14 billion with full CBAM expansion across all covered products. Pan et al. (2025) project CBAM costs of EUR 1,157 million by 2034 for China's iron, steel, and aluminium exports to the EU, with over 65% of the

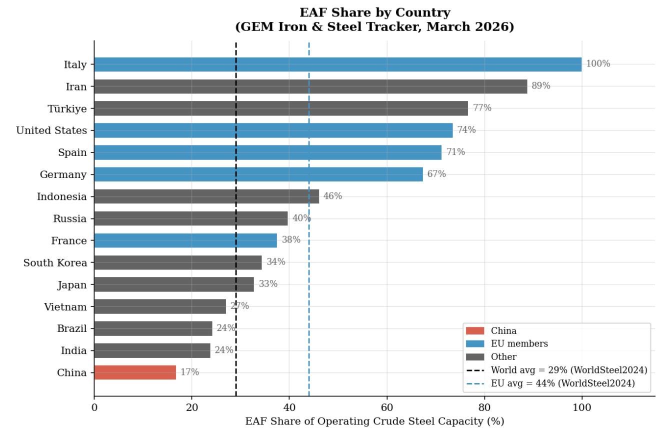

Fig. 3. EAF share of crude steel production capacity (%) by country. China's 16.9% compares with ≈29% globally and ≈44% in the EU, and >70% in the United States (World Steel Association, 2024). Data: GEM Iron & Steel Tracker, March 2026 (Global Energy Monitor, 2026).

burden falling on Hebei, Jiangsu, Shanxi, Liaoning, and Zhejiang. Provincial BF-BOF capacity distribution underlying this geographic concentration is shown in Fig. 4. Actual exposure depends on export orientation; inland capacity serving domestic markets may face lower realized CBAM costs than coastal export-oriented capacity of equivalent scale.

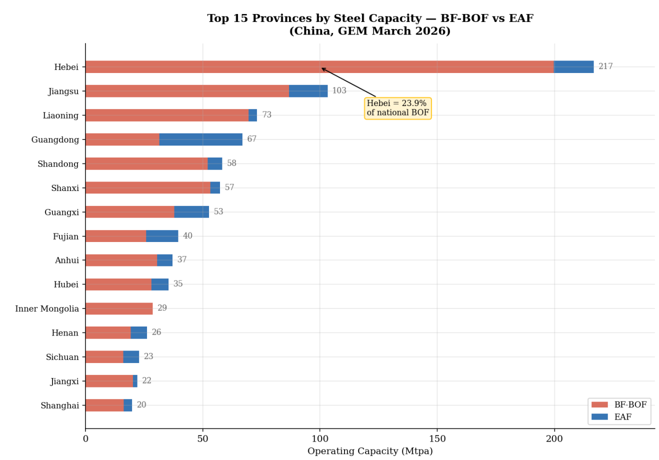

Fig. 4. Operating BOF and EAF capacity (Mtpa) by province, China. Hebei (199.7 Mtpa BOF), Jiangsu (86.8), Liaoning (69.5), Shanxi (53.2), and Shandong (52.0) collectively account for 55.2% of national BOF capacity. Data: GEM Iron & Steel Tracker, March 2026 (Global Energy Monitor, 2026).

Cement. China accounts for approximately 50–51% of global cement output (Global Cement and Concrete Association, 2024). GEM identifies 1,016 operating plants with 1,772 Mtpa cement capacity and 1,346 Mtpa clinker capacity. At the IPCC (2006) clinker emission factor of 0.825 tCO_2/t clinker (Intergovernmental Panel on Climate Change, 2006), Chinese integrated plants emit approximately 0.627 tCO_2/t of finished cement. Direct CBAM exposure from cement exports to the EU is currently limited but would enlarge substantially under any downstream construction materials expansion. The absence of large direct flows does not eliminate exposure: cement is embedded in higher-value construction products (precast elements, ready-mix concrete) whose Scope 3 emissions could fall within future CBAM reviews, and China's cement-export profile is heavily oriented toward emerging-market construction markets where any climate-club expansion would have second-order trade implications.

Aluminium. Aluminium is excluded from quantitative analysis in this paper due to the absence of GEM unit-level smelting data for China; the qualitative profile, however, merits sectoral treatment because the structural features differ materially from those of steel and cement.

China produces approximately 57% of global primary aluminium, with two structurally distinct production profiles defined by power-source mix. Coal-dominant smelting concentrated in Xinjiang, Inner Mongolia, and Shandong typically embodies 16–20 tCO_2/t Al, dominated by Scope 2 indirect emissions from coal-fired captive power. Hydropower-based smelting in Yunnan and Sichuan typically embodies 3–4 tCO_2/t Al, an order of magnitude lower. The national weighted average lies in the 13–15 tCO_2/t Al range, roughly 2× the EU average (≈6.5 tCO_2/t Al under current grid mix).

Direct CBAM exposure for primary aluminium is moderate—Chinese primary aluminium exports to the EU are constrained by EU anti-dumping measures and capacity-utilisation constraints—but exposure through semi-fabricated products (extrusions, plates, foil) is substantial, with 2024 trade volumes in these categories exceeding 2 Mt to the EU. The CBAM Annex I currently covers selected primary and semi-fabricated aluminium products; the 2026 review is expected to consider expansion to downstream aluminium goods (automotive parts, packaging), which would dramatically increase exposure.

The strategic implication is that aluminium's CBAM cost is highly dispatchable across provinces: contracting EU-bound smelting volumes to hydropower-rich provinces would lower per-tonne liability by EUR 10–15/t per tCO_2 of intensity reduction at $P_{\mathrm{EU}} =$ EUR 65/tCO_2, a far larger differential than the steel BF-BOF/EAF gap. Aluminium thus represents the sector with the largest potential for short-term CBAM-cost arbitrage through within-China relocation, distinct from the deeper technology-replacement requirements in steel.

B. CBAM Cost Quantification

Applying Eq. 1 with emission factors from Section III-B, and with steel and cement carrying $P_{\mathrm{CN}} \approx$ EUR 10–13/tCO_2 following their 2024 CN-ETS inclusion, net CBAM liability per tonne is given in Table II and visualized in Fig. 5.

Aggregate fiscal exposure. Aggregate CBAM cost depends jointly on per-tonne liability, EU-bound export volumes by sector, and the EU ETS price scenario. Under the current pilot scope and current carbon-price gap, Li et al. (2026) and Pan et al. (2025) jointly bound the annual fiscal exposure at EUR 0.78–1.16 billion, with the latter rising to EUR 1.16 billion by 2034 in their projection. Under the post-2026-review scope expansion, Li et al. (2026) project a ceiling of EUR 11.14 billion. We treat these figures as outer

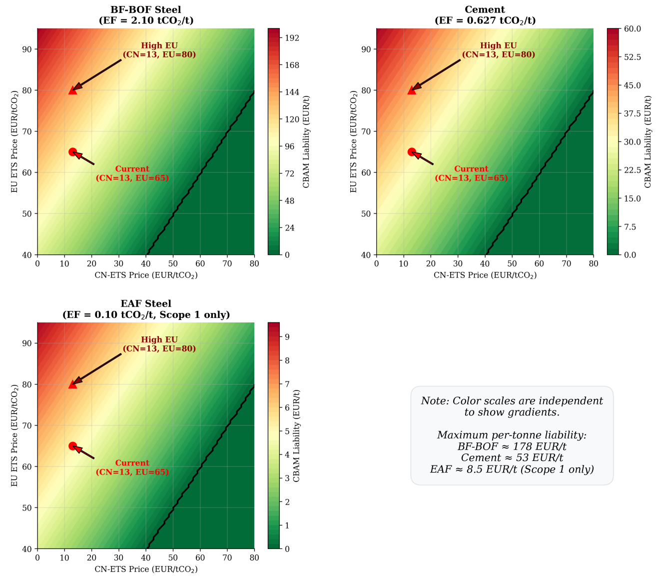

Fig. 5. Heatmap of net CBAM liability (EUR/t) across EU ETS price scenarios and CN-ETS coverage levels for BF-BOF steel, EAF steel, and cement. Derived from Table II.

bounds rather than central estimates, given heterogeneity in underlying methodology: Pan et al. (2025) construct estimates from province-level capacity data within a dynamic IAM, while Li et al. (2026) employ static GMRIO that captures sectoral input–output linkages including indirect downstream exposure absent from the former.

Provincial burden distribution. The provincial allocation of this fiscal exposure is highly skewed. Pan et al. (2025) attribute over 65% of projected CBAM costs to five provinces—Hebei, Jiangsu, Shanxi, Liaoning, and Zhejiang—broadly tracking BF-BOF capacity distribution (Fig. 4). Hebei alone bears an estimated 23% of the national burden under their methodology. This concentration creates an asymmetric political economy of reform: a small set of provincial governments would bear most of the visible cost, but national-level fiscal benefits from CN-ETS revenue retention accrue centrally, generating intergovernmental coordination challenges that the analytical framework in Section II-B treats as an extension of Putnam's two-level logic to subnational federal-style bargaining.

Sensitivity to emission factors. The base case uses point estimates for emission factors (BF-BOF 2.10, EAF Scope 1 only 0.10, cement 0.627 tCO_2/t). Plausible ranges around these, drawn from International Energy Agency (2023), GEM unit-level data, and Intergovernmental Panel on Climate Change (2006), are: BF-BOF 1.95–2.30, EAF Scope 1 0.08–0.12, cement 0.58–0.68 tCO_2/t. Because Eq. 1 is linear in EF_i, the per-tonne liability scales proportionally with the emission factor: a ±10% deviation in BF-BOF intensity translates into a ±10% deviation in BF-BOF liability, and similarly for cement and EAF. The qualitative ranking BF-BOF $\gg$ cement $>$ EAF (Scope 1) is preserved across the entire plausible range, since the gaps between sectoral intensities exceed the within-sector uncertainty by an order

TABLE II

NET CBAM LIABILITY PER TONNE OF PRODUCT BY CN-ETS COVERAGE AND PRICE SCENARIO (EUR/T)

CN-ETS Scenario	EU ETS = EUR 65/tCO_2 BF-BOF	EAF	Cement	EU ETS = EUR 80/tCO_2 BF-BOF	EAF	Cement
Not covered (pre-2024 baseline)	136.5	6.5	40.8	168.0	8.0	50.2
Covered, EUR 13/tCO_2 (*current*)	109.2	5.2	32.6	140.7	6.7	42.1
Covered, EUR 40/tCO_2	52.5	2.5	15.7	84.0	4.0	25.1
Covered, EUR 60/tCO_2	10.5	0.5	3.1	42.0	2.0	12.5
Full parity (EU ETS level)	0.0	0.0	0.0	0.0	0.0	0.0

Note: Emission factors: BF-BOF 2.10 tCO_2/t (International Energy Agency, 2023); EAF **0.10 tCO_2/t Scope 1 only** (CBAM-applicable; Scope 2 electricity emissions excluded per Regulation EU 2023/956; total EAF intensity 0.44 tCO_2/t for non-CBAM reference; International Energy Agency 2023; Ministry of Ecology and Environment of China 2023); Cement 0.627 tCO_2/t (GEM clinker ratio 0.760 × IPCC 0.825 tCO_2/t clinker; Intergovernmental Panel on Climate Change 2006). "Pre-2024 baseline" row assumes $P_{\mathrm{CN}} = 0$; "current" row reflects CN-ETS entry of steel and cement in 2024 at EUR 13/tCO_2. Figures rounded to one decimal place.

of magnitude. The provincial grid-factor question, which matters for any future Scope 2 extension, is treated separately in Section V-D.

C. Carbon Price Gap and Convergence Scenarios

As of 2024–2025, the EU ETS price has traded at EUR 50–75/tCO_2 vs. EUR 10–13/tCO_2 for CN-ETS. Three convergence scenarios toward full CBAM offset are illustrated in Fig. 6. Li et al. (2026) find that China's ETS helps reduce 30–60% of CBAM-induced export losses, with the offset magnitude positively correlated with CN-ETS price levels—underscoring the strategic value of domestic carbon pricing reform as a CBAM mitigation instrument.

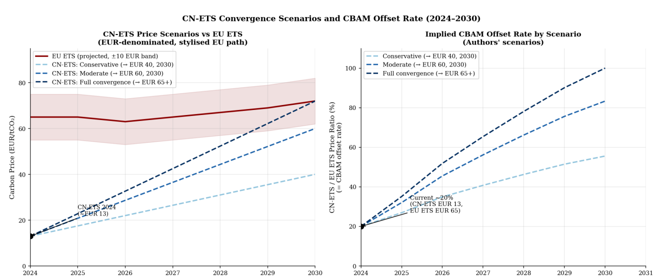

Fig. 6. CN-ETS price convergence scenarios toward EU ETS level, 2024–2035. Conservative: EUR 40/tCO_2 by 2030; Moderate: EUR 60/tCO_2 by 2030; Full convergence: price parity with EU ETS. Authors' scenarios.

Counterfactual: reform without CBAM. Attribution of CN-ETS reforms to CBAM pressure requires a counterfactual. Two observations bound it. First, CN-ETS reform announcements predate CBAM's adoption: the national ETS was announced in 2017, launched in 2021, and the dual-carbon goals (peak by 2030, neutrality by 2060) were formalised in September 2020—all before CBAM's October 2023 transitional-phase entry. Second, however, the timing of the announced 2024 sectoral expansion (steel, cement) and the 2027 absolute-cap target is closely calibrated to CBAM's definitive-phase entry in January 2026. The most defensible reading—consistent with the framework in Section II-B and with Dai and Pollitt (2025)—is that CBAM functioned as an accelerator and sequencing constraint on a reform trajectory that would have proceeded more gradually in its absence, rather than as the originating cause of reform. This interpretation acknowledges policy diffusion as multi-determined and avoids the over-attribution that single-instrument analyses can encourage.

Policy diffusion timeline. The reform trajectory traces as follows:

- **2013**: Pilot ETS programmes launched in Beijing, Shanghai, Guangdong, Tianjin, Shenzhen, Hubei, and Chongqing.
- **2017**: NDRC announces national ETS in the power sector.
- **September 2020**: Xi Jinping announces dual-carbon goals at the UN General Assembly.
- **February 2021**: National CN-ETS launched, covering power generation only (Ministry of Ecology and Environment of China, 2021).
- **October 2023**: EU CBAM transitional phase begins.
- **2024**: Steel and cement enter CN-ETS; CCER market reactivated.
- **January 2026**: EU CBAM definitive phase enters force.
- **2025–26 (planned)**: Aluminium enters CN-ETS.
- **2027 (planned)**: CN-ETS transition from intensity-based to absolute cap; MEE auctioning pilot.
- **January 2027**: UK CBAM enters force (HM Revenue and Customs, 2026).
- **2030–35**: Convergence trajectory toward EU ETS price level conditional on cap stringency and offset policy.

The clustering of activity in 2024–27—bracketing CBAM's definitive-phase entry—is the empirical signature most consistent with the signalling-channel mechanism (Section IV-A). It is not, however, evidence of single-cause CBAM-driven reform: the dual-carbon timeline would have produced sectoral expansion in roughly the same window even absent EU pressure, though probably with more relaxed initial caps and slower auctioning.

D. Robustness Checks

The point estimates above rest on parameter assumptions that warrant sensitivity testing in three directions: alternative grid factors, EU ETS price volatility, and limits on causal identification.

Alternative grid factors and the Scope 2 question. The MEE national grid emission factor (0.581 $kgCO_2$/kWh) used in Section III-B masks substantial regional heterogeneity. Provincial grid factors range from approximately 0.30 $kgCO_2$/kWh in hydropower-dominant Yunnan and Sichuan to 0.85+ $kgCO_2$/kWh in coal-dominant North China and Northwest grid regions. Under the current Regulation EU 2023/956 scope, this regional variation does *not* affect steel CBAM liability, since Scope 2 is excluded. It does, however, affect two adjacent quantities: (i) the technology-comparison total intensity for EAF (0.27 in hydropower regions vs. 0.59 tCO_2/t in coal-dominant regions), which informs the policy signalling channel even where it does not directly enter liability; and (ii) the latent exposure of Chinese aluminium and steel under any future CBAM revision that brings Scope 2 into scope. The latter is particularly consequential for aluminium: regional grid-factor substitution shifts aluminium intensity from $\approx$13 tCO_2/t (national average) to $\approx$3.5 tCO_2/t (hydropower) or $\approx$18 tCO_2/t (coal), bracketing the entire EU–China gap. The implication is that any future CBAM expansion to Scope 2 for steel and aluminium would generate substantial within-China heterogeneity in exposure—a feature that itself becomes a CN-ETS design parameter.

EU ETS price volatility. The base scenarios use point estimates of EUR 65 and EUR 80/tCO_2 for P_{EU}. Historical 2024–25 EU ETS prices ranged from EUR 50/tCO_2 (mid-2024 trough) to EUR 85/tCO_2 (early-2023 peak), with annualised volatility of approximately 28%. At P_{EU} = EUR 50, BF-BOF liability under current P_{CN} falls to EUR 77.7/t (vs. EUR 140.7 at EUR 80); at P_{EU} = EUR 100, it rises to EUR 182.7/t. The roughly 2$\times$ range across plausible price scenarios reinforces that CBAM exposure is materially uncertain even before considering CN-ETS reform dynamics, and underscores the value of the deduction clause in dampening this volatility from the perspective of Chinese exporters: any increase in P_{EU} that lifts CBAM liability simultaneously raises the marginal value of P_{CN} to MEE, partially internalising the volatility.

Causal identification caveats and selection effects. As noted in Section III, the correlation between CBAM implementation and CN-ETS reform acceleration is consistent with the framework in Section II-E but does not constitute formal causal identification. Counterfactual evidence—e.g., a comparable jurisdiction subject to CBAM exposure but without independent decarbonisation goals—does not exist at sufficient scale to support difference-in-differences estimation. Findings are therefore interpreted as theoretically informed associations, not demonstrated causal effects. A second selection issue arises in the CBAM-exposure analysis itself: the model considers EU-bound exports rather than total Chinese steel and cement output, implicitly assuming that EU-bound product mix matches national-average emissions intensity. If EU-bound exporters disproportionately use cleaner production technologies (a plausible response to CBAM signalling), our point estimates would over-state actual exposure; we therefore treat the figures here as upper bounds in this respect. Finally, Pan et al.'s (2025) provincial allocation uses capacity shares as a proxy for trade-flow shares, which may misallocate exposure if certain provinces specialise disproportionately in domestic-market or non-EU export production. We flag these as material limitations and recommend that future work integrate customs-level export microdata to disentangle the production-mix from the destination-mix effects.

VI. Industrial and Policy Response

A. Steel Adjustment Pathways

China's steel sector presents the most acute CBAM exposure, combining the world's highest absolute production volume with an emissions-intensive technology mix geographically concentrated in provinces where decarbonization pathways are structurally constrained (Fig. 4). The EUR 130.0/t per-tonne CBAM liability differential between BF-BOF and EAF at EUR 65/tCO_2 (pre-2024 baseline; Table II) defines the upper bound of within-firm adjustment space and can be partially neutralized through three distinct pathways, each with different time horizons and political feasibility.

Product switching for EU-bound lines. Dedicated EAF capacity serving CBAM-liable exports is the most direct response to the intensity differential. Coastal provinces with established scrap collection networks and port logistics—Jiangsu, Guangdong, Sichuan—are the natural first movers (Chen et al., 2023). The constraint here is not technology but feedstock: EAF requires steel scrap, which is in short supply domestically (discussed below).

EAF capacity expansion. Beyond product switching within existing firms, system-wide EAF capacity expansion would reduce the weighted-average emissions intensity of China's steel fleet. This is the slowest pathway because it requires not only new EAF investment but also the upstream scrap supply chain and downstream product adjustment (EAF excels at construction steel; specialty steel still requires integrated routes).

Process adjustment within BF-BOF. Increased scrap charging into basic oxygen furnaces and partial hydrogen injection at the blast furnace stage can reduce per-tonne intensity by an estimated 10–20% short of full technology replacement. This pathway preserves existing capital stock and is therefore the most politically attractive in BF-BOF-heavy provinces—it allows incremental decarbonization without stranded-asset losses.

The scrap supply constraint binds all three pathways. China's steel fleet is relatively young, reflecting the rapid capacity build-out of the 2000s. The buildings, bridges, and rail networks constructed from that steel have not yet reached end-of-life, so domestic scrap generation remains structurally limited. Scrap availability is projected to rise substantially through 2030–2035 as early 2000s infrastructure begins reaching recovery (International Energy Agency, 2023), enabling a medium-term decarbonization wave that CBAM may accelerate by rewarding early EAF investment—consistent with, though not demonstrated by, the fiscal incentive channel in Section IV-A.

Geographic concentration amplifies political constraints. The provincial concentration of BF-BOF capacity is not incidental to adjustment difficulty—it is the central feature. Hebei alone holds 199.7 Mtpa of BF-BOF capacity (23.9% of national total), and the top five provinces account for 55.2%. These same provinces house the coking coal supply chains and steel-dependent labour markets that make rapid BF-BOF retirement politically costly at the subnational level. Chen et al. (2023) project that CCUS deployment will accordingly concentrate in Hebei, Shandong, Liaoning, and Jiangsu, while EAF expansion concentrates elsewhere—a geographic bifurcation of decarbonization strategy that maps directly onto the political economy constraints discussed in Section VI-D.

B. *Cement Decarbonization Constraints*

Cement decarbonization is constrained by the thermodynamic irreducibility of calcination: ≈63% of direct emissions arise from $CaCO_3 \rightarrow CaO + CO_2$, which cannot be eliminated through fuel switching or energy efficiency alone. GEM identifies 1,016 operating plants with 1,772 Mtpa cement capacity; 782 are fully integrated and 214 are grinding-only. The clinker-to-cement ratio of 0.760 exceeds the global average of 0.73 (Global Cement and Concrete Association, 2024). Translating the cost exposures modelled in Section V-B into investment imperatives, CCS deployment status and CBAM-implied breakeven economics are shown in Fig. 7.

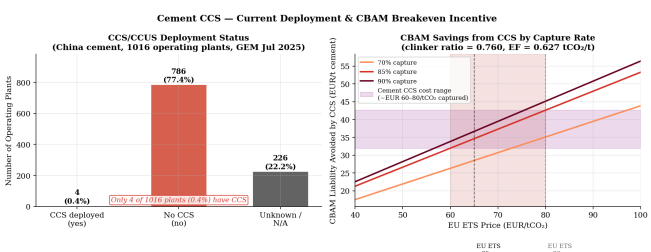

Fig. 7. Cement sector CCS deployment in China and CBAM-implied CCS breakeven cost. Of 1,016 GEM-tracked operating plants, only 4 report active CCS/CCUS (<0.4%). Breakeven assumes 90% CO_2 capture rate (Zhong et al., 2024).

The CBAM cost floor (Section V-B) provides a partial economic signal for CCS investment. With a post-combustion capture rate $\eta = 0.90$ (Zhong et al., 2024), the CBAM saving per tonne of cement from deploying CCS is:

$$\Delta_{\text{CBAM}} = EF_{\text{cement}} \times \eta \times P_{\text{EU}} = 0.627 \times 0.90 \times 65 \approx \text{EUR } 36.7/\text{t cement} \tag{2}$$

CCS becomes economically viable when this saving covers operating costs. At the prevailing CCS cost of EUR 60–80/tCO_2 captured (= EUR 33.7–45.1/t cement), the breakeven EU ETS price is approximately EUR 65–80/tCO_2—placing current EU ETS levels at the lower bound of CCS viability for cement.

The technical floor for clinker content in ordinary Portland cement is 0.65–0.70. China's primary SCMs—fly ash and blast furnace slag—are byproducts of sectors undergoing structural decline, constraining the clinker substitution pathway and reinforcing the long-run indispensability of CCS.

Why deployment remains marginal despite favourable economics. That only 4 of 1,016 GEM-tracked plants report active CCS/CCUS, even as the breakeven calculation in Eq. 2 approaches viability at current EU ETS prices, points to barriers beyond the per-tonne capture cost itself. Three are particularly binding in the Chinese context. *First*, the carbon price signal that drives the breakeven calculation operates at P_{EU}, not at P_{CN}; domestic cement producers face a CN-ETS price of EUR 10–13/tCO_2, well below the threshold at which CCS becomes economically attractive on domestic compliance grounds alone. The CBAM-implied incentive applies only to EU-bound cement exports, which currently represent a small share of Chinese cement output. *Second*, CCS deployment requires CO_2 transport and storage infrastructure that China has not yet built at scale. Suitable geological storage sites are concentrated in the Songliao and Ordos basins (Northeast and Northwest China), while high-emission cement and steel facilities concentrate in Hebei, Shandong, and Jiangsu—a geographic mismatch that requires substantial pipeline investment to resolve. *Third*, the commercial model for CCS remains unclear: in the absence of dedicated subsidies or a sufficiently high domestic carbon price, individual firms have no incentive to bear the first-mover infrastructure costs. The result is a coordination failure that is unlikely to resolve through CBAM pressure alone, even as the per-tonne economics tighten.

C. *CN-ETS Reform: Before and After CBAM*

China launched its national ETS in February 2021, initially covering only the power generation sector (≈40% of total CO_2 emissions) (Ministry of Ecology and Environment of China, 2021). The intensity-based allocation mechanism does not constrain absolute emissions, and allowance prices have traded in the range of CNY 40–105/tCO_2 (EUR 5–13/tCO_2)— far below EU ETS levels.

MEE has since announced three central reforms aligned with CBAM's definitive phase: (i) transition from intensity-based to absolute emissions caps by 2027; (ii) sectoral expansion, with steel and cement incorporated into CN-ETS in 2024 and aluminium targeted for inclusion in 2025–2026; and (iii) introduction of allowance auctioning to replace free allocation.

The three reforms are interdependent. Absolute caps without sectoral expansion would constrain only the power sector, leaving CBAM-exposed industries outside the deduction-clause pathway. Sectoral expansion without absolute caps would extend coverage but not generate price discovery, since intensity benchmarking allows allowance volumes to expand with output and dampens upward price pressure (Dai and Pollitt, 2025; Stoerk et al., 2019). Auctioning without binding caps would convert free allowances into revenue but leave the underlying scarcity unchanged. The strategic value of the announced reform package lies in the joint effect: absolute caps create price signal, sectoral expansion extends it to CBAM-relevant industries, and auctioning converts the resulting price into realized fiscal flows that Article 9 can offset against CBAM liability.

Attribution and counterfactual. Attribution of these reforms solely to CBAM pressure requires caution. The dual carbon goals (*shuang tan*)—peak CO_2 by 2030, neutrality by 2060—were announced in September 2020, before CBAM's adoption in May 2023. CN-ETS launched in February 2021, before CBAM's transitional phase began in October 2023. The reform direction was therefore set independently. What appears to be calibrated to CBAM is the *timing* and *sequencing*: the 2024 sectoral expansion and 2027 absolute-cap target bracket CBAM's January 2026 definitive-phase entry. The most defensible reading—consistent with the analytical framework in Section II-E—is that CBAM functions as an accelerator and sequencing constraint on a reform trajectory already underway, rather than as the originating cause of reform. We claim correlation, not causation: a formal causal identification would require a counterfactual China without CBAM exposure, which does not exist.

D. Political Economy Constraints

Three structural constraints govern the pace of CN-ETS reform and industrial decarbonization. They are interlocking rather than independent, and together define the boundaries of feasible reform.

SOE dominance and the cost-benefit asymmetry. State-owned enterprises hold the majority share of capacity in steel, cement, and aluminium. The cost-benefit calculus they face is structurally asymmetric: SOEs internalize compliance costs of carbon pricing directly but capture only a fraction of the export competitiveness benefits, because a large share of their output serves domestic markets rather than EU-bound exports. This asymmetry produces a consistent institutional preference for slower reform and more generous free allocation (Nahm and Urpelainen, 2021). SOEs are not opposed to decarbonization in principle—many have public commitments aligned with the dual-carbon goals—but the bureaucratic position they advocate in inter-ministerial deliberations consistently favours gradualism over price discovery.

Energy security and regional employment dependence. Blast furnace operations in China are not standalone industrial facilities; they are embedded in regional supply chains that link coking coal mining, coke production, blast furnace operation, and downstream steel processing. Hebei, Shanxi, and Liaoning provinces have structural employment dependence on this integrated coal-steel complex—rapid retirement of BF-BOF capacity would generate employment and fiscal-revenue shocks that provincial governments are politically unwilling to absorb. The constraint is not technological (EAF is technically mature) but political-economic (the transition distributes losses and gains across actors with very different political weights).

Regional carbon intensity heterogeneity. Provincial grid emission factors range from approximately 0.30 $kgCO_2/kWh$ in hydropower-dominant Yunnan and Sichuan to over 0.85 $kgCO_2/kWh$ in coal-dominant North and Northwest China. A uniform national carbon price imposes structurally different burdens across provinces. For aluminium, the implication is particularly stark: hydropower-based smelting in Yunnan and Sichuan embodies 3–4 tCO_2/t Al, while coal-based smelting in Xinjiang, Inner Mongolia, and Shandong embodies 16–20 tCO_2/t Al. Under uniform carbon pricing, the same nominal price functions as a modest cost adjustment for Yunnan and as a sectoral threat to Xinjiang—generating distributional tensions that constrain the political feasibility of stringent absolute caps. The same logic applies, less dramatically, to steel and cement: provinces with cleaner power mixes face manageable compliance costs, while coal-dependent provinces face costs that approach the scale of their sectoral output.

Putnam's framework as constraint map, not causal model. We use the two-level games framework (Putnam, 1988) descriptively rather than predictively: the three constraints above define the boundaries of China's domestic win-set, identifying the range of reform outcomes that are politically feasible. CBAM shifts the fiscal calculus of the status quo—by attaching a measurable fiscal-leakage cost to non-reform—but does not by itself determine which outcome within the win-set materializes. Putnam's original formulation was developed for democratic states with formalized legislative ratification; China's win-set is shaped through party channels, inter-ministerial deliberation (MEE, NDRC, MIIT, MOFCOM), and SOE advisory bodies rather than legislative veto, which limits the framework's direct predictive applicability. We retain it as an analytical lens for mapping constraints, not as a forecasting tool. The cumulative implication of these constraints is that reform trajectories will be gradual, sectoral, and subject to implementation delays relative to MEE's announced timelines—particularly where they intersect SOE interests, energy security imperatives, and provincial fiscal distributional concerns simultaneously.

VII. Discussion

The empirical findings in Sections V and VI establish four observations that anchor the discussion. *First*, per-tonne CBAM liability is dominated by the 17.6:1 emissions-intensity ratio between BF-BOF and EAF steel (Table II)—technology heterogeneity, not aggregate sectoral size, is the binding parameter. *Second*, CN-ETS pricing operates as a fiscal lever through Regulation EU 2023/956's Article 9 deduction clause: each euro paid domestically is a euro not transferred to the EU budget. *Third*, the political economy of reform (Section VI-D) is geographically concentrated, with five provinces bearing the bulk of exposure while fiscal benefits accrue centrally. *Fourth*, the analytical framework we adopt imposes a structure on the China side of the analysis that warrants explicit reflection. We translate the first three observations into recommendations across three audiences—firms, policymakers, and the EU–China relationship—and use the fourth to frame the cooperation discussion.

A. Enterprise-Level Strategies for Reducing CBAM Exposure

Section V-B establishes that CBAM liability is driven by per-unit carbon intensity rather than aggregate emissions.

Firm-level response should therefore target the intensity gap directly. The implications differ substantially between steel and cement.

Steel exporters: three layered strategies. The EUR 134.0/t per-tonne differential between BF-BOF and EAF at P_{EU} = EUR 80/tCO_2 defines the upper bound of within-firm CBAM-cost arbitrage. Three pathways follow, each with different time horizons and capital intensity.

Short-term portfolio rebalancing is the cheapest and most immediate. Firms should treat CBAM exposure as a binding allocation parameter: EU-bound order books should preferentially be filled from lower-intensity capacity (existing EAF lines, scrap-rich BF-BOF operations), with high-intensity capacity redirected to domestic and non-CBAM export markets. This is not an emissions reduction strategy—global emissions are unchanged—but it is a CBAM-cost mitigation strategy entirely consistent with current Regulation EU 2023/956 scope. The corresponding limitation, which firms should anticipate, is that this strategy becomes increasingly ineffective as other major economies adopt their own border carbon adjustments (Section IV-D); the destination-substitution option narrows as CBAM-like regimes diffuse to the United Kingdom and, potentially, the United States.

Medium-term process adjustment within BF-BOF—scrap charging and partial hydrogen injection—can reduce intensity meaningfully short of full technology replacement, providing a transition path during the 2025–2030 scrap-supply constraint window. The investment threshold is moderate, and the existing BF-BOF capital stock is preserved. Firms in BF-BOF-heavy provinces (Hebei, Shanxi) face asymmetric returns to this pathway: it allows participation in decarbonization without stranded-asset losses, making it politically the most defensible response.

Long-term dedicated EAF capacity for EU-bound product lines exploits the BF-BOF/EAF intensity differential at its maximum. Coastal provinces with established scrap collection networks and port logistics—Jiangsu, Guangdong, Sichuan—are the natural candidates (Chen et al., 2023). The binding constraint is scrap supply, which is not expected to release at scale until 2030–2035 as early-2000s infrastructure reaches end-of-life. EAF strategy is therefore a 5–10 year investment decision, not a near-term response to current CBAM exposure.

Cement producers: narrower options, sharper imperatives. The thermodynamic irreducibility of calcination means that supplementary cementitious material substitution and CCS deployment are the only material decarbonization pathways. Eq. 2 establishes that CCS breakeven sits at the lower bound of current EU ETS prices, but as Section VI documents, only 4 of 1,016 GEM-tracked plants have deployed CCS—a coordination failure that the per-tonne economics alone are unlikely to resolve. For individual cement firms, the most cost-effective near-term investment is therefore not capture technology but MRV readiness: the operational distinction between EU ETS-aligned facility-specific emission factors and the punitive default values applied where MRV is absent is large enough to justify investment in ISO 14064-aligned carbon accounting independent of CCS deployment timing.

Across both sectors: documentation as no-regret investment. Sectors not yet inside CN-ETS face the maximum CBAM liability under the current price gap (Table II, "not covered" row). Firms should build MRV data infrastructure now so that domestic carbon payments are immediately offsettable once CN-ETS coverage extends to their sector. The sequencing matters: the cost of MRV readiness is fixed, but the value depends on having the documentation in place at the moment of CN-ETS entry. Firms that delay incur compounded costs through the transition.

B. Carbon Market Reform: Closing the Price Gap to Retain Revenue

Sections V-B and VI-A establish the revenue-retention logic that reframes the carbon-pricing reform question. Before CBAM, raising P_{CN} was a question of domestic distribution: who within China bears the abatement cost. After CBAM, raising P_{CN} is also a question of international fiscal allocation: each euro increase shifts an equivalent quantum of fiscal capacity from the EU budget to the Chinese budget, bounded above by P_{EU}. This dual character changes both the magnitude of the policy stakes and the coalition of domestic actors with an interest in reform.

Four reform vectors follow from this logic, each addressing a different element of the deduction mechanism.

Absolute caps as the precondition for price discovery. The announced 2027 transition from intensity-based to absolute caps is the necessary condition for sustained upward price pressure under CN-ETS. Intensity benchmarking allows allowance volumes to expand with output, dampening price signals (Dai and Pollitt, 2025; Stoerk et al., 2019); absolute caps cap aggregate emissions and therefore generate scarcity. Table II shows the fiscal stakes: reaching P_{CN} = EUR 40/tCO_2 by 2030 would reduce per-tonne CBAM liability by approximately 40% across BF-BOF, EAF, and cement at P_{EU} = EUR 80; reaching EUR 60 would reduce it by approximately 70%. Cap stringency is the key calibration parameter of the entire CBAM-CN-ETS interaction.

Sectoral expansion ahead of CBAM scope deepening. Bringing aluminium into CN-ETS in 2025–2026, on the announced timeline, captures the revenue-retention benefit before any CBAM scope review deepens aluminium exposure. The sequencing matters because each year a sector remains outside CN-ETS while CBAM-liable is a year of pure fiscal outflow—no portion of the liability is offsettable. The same logic applies to any future CBAM scope extension into downstream goods (organic chemicals, polymers, manufactured products): the fiscal cost of delayed CN-ETS coverage scales with the scope of extension.

Auctioning and reserve prices. Free allocation leaves the deduction clause working only at the margin, because the *realized* compliance cost—the cost that Article 9 actually deducts—is determined by what firms pay for allowances, not by the notional carbon price. Introducing a gradually rising auction share, with a reserve price calibrated to the threshold where CN-ETS payments begin materially offsetting CBAM costs, would convert revenue retention from a notional price signal into a realized fiscal instrument. The 2027 auctioning pilot is the operational vehicle for this conversion.

Revenue recycling and the central-provincial bargain. The fiscal proceeds, once realized, should be earmarked for the technology and readiness gaps identified in Sections VI-A–VI: EAF capacity and scrap infrastructure subsidies, renewable-electricity incentives for aluminium relocation, enterprise MRV grants, and—critically—transfers to high-exposure provinces. The political economy constraints in Section VI-D make this last element non-optional. SOE-dominated steel and cement firms in Hebei, Shanxi, and Liaoning bear the visible costs of any cap-tightening regime; the fiscal benefits of revenue retention accrue to the central government through MEE-administered channels. This intergovernmental asymmetry is a binding constraint on cap stringency that no central directive can dissolve. A reform path that does not include explicit compensation mechanisms for high-exposure provinces will encounter implementation resistance regardless of central ambition—and that resistance will materialize through the inter-ministerial channels (NDRC, MIIT, MOFCOM) that the political economy literature on Chinese authoritarian governance has long identified (Chen and Lo, 2025; Feng, 2021).

Sequencing and feasibility. The four vectors above are not independent reform initiatives but components of a single sequenced package. Absolute caps without sectoral expansion would constrain only the power sector; sectoral expansion without absolute caps would extend coverage but not generate price discovery; auctioning without binding caps would convert free allowances into revenue but leave underlying scarcity unchanged; revenue recycling without the preceding three is a transfer mechanism without a source. The political feasibility of the package depends on simultaneous movement, which in turn depends on the institutional balance between MEE (advocating ambition), NDRC (managing growth implications), and provincial governments (absorbing transition costs).

C. EU–China Cooperation: From Coercive Pressure to Market Alignment

The carbon price gap (Section V-B) and the 30–60% offset ceiling (Li et al., 2026) together define the fiscal stakes of the EU–China carbon-policy interaction. Cooperation can raise that ceiling through a phased pathway, but cooperation requires a framing shift that we name here explicitly, as it conditions the discussion that follows.

Reframing the agency question. The dominant analytical framing of CBAM and China treats the EU as agenda-setter and China as respondent: an external incentive structure produces an internal Chinese reform response. This framing is embedded in the analytical apparatus we adopt—Putnam's two-level games framework presupposes external pressure transmitting into domestic politics—and we have applied it throughout the paper. But the framing is incomplete in a way that bears directly on cooperation prospects. China's dual-carbon goals (announced September 2020) predate CBAM's adoption (May 2023). CN-ETS launched in February 2021, before CBAM's transitional phase began in October 2023. China is the core manufacturer of the EU's green transition—photovoltaic modules, batteries, electric vehicles, wind turbine components—and the EU's 2030 climate ambition is structurally dependent on Chinese industrial capacity. China is also a principal voice in the G77+China coalition, defending the principle of common-but-differentiated responsibilities (CBDR) and pressing for climate-finance commitments from developed nations. The empirical pattern in Section VI is therefore more accurately described as CBAM *accelerating and sequencing* a reform trajectory that China is already pursuing on its own terms; CBAM does not initiate that trajectory. This distinction has operational consequences: a cooperation framework premised on China-as-respondent is unlikely to elicit the genuine mutual recognition that effective MRV equivalence requires, because it asks China to accept a subordinate role in rule-making it has independent grounds to contest.

Three cooperation pathways under a co-shaper framing. With the framing shift in place, three sequential cooperation opportunities become tractable.

Short term (2025–27): mutual recognition of CN-ETS payments as CBAM offsets at facility level. The current 30–60% offset rate (Li et al., 2026) reflects in part the absence of formal MRV equivalence: Chinese firms can declare CN-ETS payments, but verification procedures and accreditation pathways remain incomplete. A bilateral recognition agreement could lift the realized offset rate toward 80–90% conditional on third-party-verifier accreditation and harmonized emission-factor methodologies. This is the cooperation step with the largest immediate fiscal return for China and the lowest political cost for the EU—it changes the operation of existing rules without altering the underlying regulatory architecture. It also generates institutional infrastructure (verified Chinese MRV data, accredited verifiers operating to ISO 14064 standards) that has value beyond the EU–China dyad.

Medium term (2027–30): MRV harmonization at scale. Building on facility-level recognition, full MRV harmonization through joint emission-factor methodologies and mutual recognition of accredited verifiers would allow Chinese installation-level data to replace CBAM default values at scale. This requires sustained technical dialogue between MEE and DG TAXUD, modelled on existing Sino-European environmental cooperation but with substantially deeper operational integration. It is also the operational precondition for any future ETS linkage.

Long term (post-2030): price-gap convergence and the linkage question. If the price-gap convergence trajectory in Section V narrows the gap sufficiently, one-way market linkage on the model of EU–Switzerland ETS linkage becomes technically conceivable. Linkage is politically efficient—it forecloses the entire CBAM-on-China question by aligning prices directly—but it requires substantial regulatory convergence and a degree of mutual trust in domestic enforcement that does not currently exist. Linkage should be understood as a long-horizon aspiration that conditions the medium-term harmonization agenda, not as a near-term policy option.

The transferable template. The CN-ETS–CBAM interaction also offers a transferable template that extends the significance of this analysis beyond the bilateral relation. The revenue-retention logic gives other developing-country ETS jurisdictions—South Korea (operational since 2015), Indonesia (launched 2023), Thailand (under development), and potentially India (proposed Carbon Credit Trading Scheme)—a fiscal rather than purely environmental rationale for domestic carbon pricing reform. Environmental arguments alone have historically struggled to generate sustained political momentum in emerging economies, where development and energy access concerns take precedence. CBAM's deduction structure attaches a tangible fiscal cost to non-action that environmental arguments do not. To the extent this template diffuses, the significance of CBAM extends beyond the EU–China bilateral interaction into broader carbon-policy architecture across Asia and, eventually, the Global South.

Limits of the cooperation framing. The cooperation pathways above are not guaranteed. They depend on a baseline of bilateral trust that the broader trade and technology environment may not provide. The expansion of border carbon adjustment regimes beyond the EU (Section IV-D) creates parallel pressures that the deduction-based EU framework cannot internalize. And the fundamental tension between China's official position—that CBAM is a unilateral measure inconsistent with WTO non-discrimination and CBDR principles—and the operational reality of CN-ETS adjustment is unlikely to resolve in either direction. Cooperation on MRV is compatible with continued contestation of the underlying regulatory legitimacy; in fact, that combination—operational adjustment without rhetorical concession—is the empirical pattern we observe. Effective cooperation should be designed around this pattern, not in opposition to it.

VIII. Conclusion

This paper has examined how the EU's Carbon Border Adjustment Mechanism reshapes China's carbon pricing architecture and sectoral decarbonization pathways. We approached this question through a three-stage framework: plant-level data on 458 steel and 1,016 cement facilities; a CBAM liability model derived directly from Regulation EU 2023/956 and applied across two EU ETS price scenarios and five CN-ETS price states; and a political-economy reading informed by the two-level games tradition but adapted to the authoritarian-bureaucratic context. The substantive contribution is a three-channel transmission framework—direct cost, revenue retention, signalling—through which CBAM's external design produces internal pressure on Chinese policy.

Three findings carry the argument.

First, per-tonne CBAM liability is dominated by technology heterogeneity rather than aggregate sector exposure. At P_{EU} = EUR 80, BF-BOF steel faces a net liability of EUR 140.7/t against EUR 6.7/t for EAF, a 17.6:1 ratio that maps emissions performance directly onto trade competitiveness. The practical implication is that firm-level and policy-level responses should target the intensity gap directly. Sector-aggregate exposure estimates—useful as headline figures—obscure the within-sector technology distribution that drives the actual competitive adjustment.

Second, the deduction clause in Regulation EU 2023/956 transforms domestic carbon pricing from a purely environmental instrument into a fiscal substitution mechanism. Each euro of CN-ETS price retained in China is a euro not transferred to the EU budget, bounded above by P_{EU}. This reframing aligns the political economy of CN-ETS reform with treasury rather than environmental ministry incentives—a non-trivial reorientation, because it expands the coalition of domestic actors with a direct fiscal interest in reform. The mechanism is most powerful at the margin: each successive euro of P_{CN} produces an identical fiscal capture until parity, at which point CBAM liability falls to zero.

Third, the empirical timing pattern of Chinese carbon-pricing reform—steel and cement entering CN-ETS in 2024, the announced 2027 absolute-cap transition, sectoral expansion to aluminium in 2025–2026—is consistent with, but does not demonstrate, CBAM-induced reform acceleration. China's dual-carbon goals provide independent momentum that predates CBAM by more than two years. The most defensible reading is that CBAM functions as accelerator and sequencing constraint on a reform trajectory already underway, not as the originating cause. We have been deliberate throughout the paper in claiming correlation rather than causation, and in treating the analytical framework as a constraint map rather than a causal model.

Policy implications differ by audience. For Chinese firms, the per-unit intensity gap is the binding parameter, and the appropriate response sequence is short-term portfolio rebalancing, medium-term BF-BOF process adjustment, and long-term dedicated EAF investment. MRV readiness is a no-regret investment across all sectors and time horizons. For the Chinese government, accelerating CN-ETS price discovery through absolute caps, sectoral expansion, auctioning, and revenue recycling converts CBAM exposure into domestic fiscal capacity—but requires explicit central-provincial compensation mechanisms to overcome the geographic concentration of adjustment costs. For the EU–China relationship, mutual recognition of MRV equivalence offers the largest near-term fiscal gain at lowest political cost, with deeper harmonization and eventual market linkage as medium- and long-term aspirations.

The transferable element is the revenue-retention logic itself, which gives other developing-country ETS jurisdictions (South Korea, Indonesia, Thailand, India) a fiscal rather than purely environmental rationale for domestic carbon pricing reform. Environmental arguments alone have historically struggled to generate sustained political momentum in emerging economies where development and energy access concerns take precedence. The deduction structure attaches a tangible fiscal cost to non-action. To the extent this template diffuses, the significance of CBAM extends well beyond the EU–China bilateral interaction.

The study has three principal limitations that bound the generality of these findings.

First, aluminium is excluded from quantitative analysis due to the absence of plant-level smelting data comparable to the GEM trackers used for steel and cement. This is a non-trivial omission: aluminium represents a substantial share of latent CBAM exposure, particularly under any future Scope 2 extension, and the within-China heterogeneity between hydropower-based smelting (Yunnan, Sichuan) and coal-based smelting (Xinjiang, Inner Mongolia) creates a regional-arbitrage dynamic structurally distinct from steel. We discuss aluminium qualitatively in Section VI-A but cannot quantify the corresponding liabilities with the same precision.

Second, provincial cost allocations rely on capacity shares as a proxy for trade-flow shares. If EU-bound exporters disproportionately use cleaner production technologies—a plausible response to CBAM signalling—our point estimates would over-state realized exposure relative to capacity-weighted projections. Customs-level export microdata, which is not publicly available at the disaggregation we would require, would allow disentangling production-mix from destination-mix effects.

Third, and most consequentially, the correlation between CBAM implementation and CN-ETS reform acceleration is consistent with the analytical framework but does not constitute formal causal identification. A formal causal claim would require a counterfactual—a comparable jurisdiction subject to CBAM exposure but without independent decarbonization goals—which does not exist at sufficient scale to support quasi-experimental estimation. We have therefore interpreted the empirical pattern as theoretically informed association rather than demonstrated effect, and we want to flag a related framework limitation: the two-level games apparatus we adopt presupposes external pressure transmitting into domestic policy. This presupposition fits the mechanism we trace, but it understates the degree to which Chinese climate policy is shaped by domestic agenda-setting independent of external pressure. The empirical signature we observe—reform announcements predating CBAM, sequencing calibrated to CBAM, rhetorical contestation of CBAM combined with operational adjustment to it—is consistent with multiple causal stories that our framework cannot fully distinguish.

Three directions for future research follow.

First, customs-level microdata on EU-bound exports would allow disentangling production-mix from destination-mix effects that capacity-share proxies cannot resolve. Joint research with the General Administration of Customs of the People's Republic of China, or use of mirror trade statistics from Eurostat, could substantially sharpen the empirical foundation of the literature.

Second, comparative analysis of CBAM-exposed jurisdictions with and without independent decarbonization trajectories—candidate cases include Türkiye, India, and Brazil, each with different combinations of CBAM exposure and domestic carbon-policy ambition—would provide the closest available approximation to counterfactual identification. This is the most direct path to strengthening the causal inference we have deliberately avoided.

Third, and most fundamentally, a theoretical framework for non-Western agency in global climate governance, in which countries like China are treated as co-shapers rather than respondents, remains an open project. The CBAM–China interaction is particularly well-placed to inform that project, because it combines a quantifiable fiscal mechanism (the deduction clause), an observable adjustment process (CN-ETS reform), and a rhetorically contested legitimacy that no purely respondent-framing can explain. We have not constructed such a framework here, but we have tried to identify the empirical features it would need to accommodate. The CN-ETS–CBAM interaction is a productive site for developing it.

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